

Nicholas Tividad

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Education

The University of Nottingham | MSc Computer Science (Artificial Intelligence)

2025-Present

Predicted Grade: Distinction

Relevant Modules: Programming, Databases & Software, Designing Intelligent Agents, Data Science with Machine Learning.

The University of Sheffield | BEng Civil Engineering with a Year in Industry

2021-2025

Grade: 2:1

Relevant modules: Further Civil Engineering Mathematics & Computing (75% in Programming)

Altrincham Grammar School for Boys | Secondary School | Sixth Form

2014-2021

A-Levels: Physics (A), Mathematics (B), Computer Science (B)

Technical Projects

Hospital Management System

Oct 2025 – Dec 2025

- Developed a role-based hospital management web application in PHP & SQL with separate admin & doctor dashboards
- Designed and populated a relational database with 8 linked tables, connected using Docker and a LAMP stack.
- Implemented a login & session management in PHP, using prepared SQL statements to handle user authentication
- Constructed an audit log that automatically tracks every insert, update, and delete made by any user across the system.
- Created a parking permit workflow, where doctors could submit requests, admins can approve/reject them with reason.

Natural Selection Simulation

Mar 2026 – May 2026

- Built a Python natural selection simulation in Tkinter where blob agents forage, evolve and compete across generations.
- Designed an energy trade-off system where movement cost scales with speed and sense range, driving selective pressure.
- Implemented mutation, inheritance and reproduction logic to evolve agent traits across 50 generations per run.
- Automated 10 repetitions per condition with deterministic seeding to test how food scarcity affects trait evolution.
- Analysed final-generation outcomes in Pandas with ANOVA and Tukey HSD tests, visualising results in Matplotlib.

NBA Draft Prediction with Machine Learning

Feb 2026 – May 2026

- Built a Python ML pipeline predicting draft round for NBA players using only pre-draft combine measurements.
- Combined two tables from a public NBA SQLite database with SQLite3 and Pandas to assemble a 972-player dataset.
- Compared 4 Scikit-Learn classifiers (Logistic regression, Random Forest, SVM with KBF kernel, Gradient boosting) on a stratified 80/20 train-test split.
- Fitted StandardScaler on training data only to prevent data leakage, evaluating models on ROC-AUC, F1 and accuracy.
- Performed EDA with histograms, box plots and a correlation heatmap (Seaborn, Matplotlib), uncovering multicollinearity above 0.9 in physical features.

Experience

Structural Engineer Placement – Adept C&S Consulting Engineers Ltd.

Sep 2024 – Aug 2025

- Contributed to 10+ projects across educational, commercial, residential, healthcare refurbishments and extensions.
- Produced design outputs including structural models, drawings & calculations to support planning & construction stages.
- Investigated existing structures from site assessments, document reviews & intrusive surveys to inform design decisions.
- Coordinated designs with architects, M&E consultants, project managers & clients to deliver cost-effective solutions.

Crew Member – Lane 7 Altrincham

Jun 2024 – Present

- Improved service speed during busy periods by effectively keeping track of customer orders & determining priority.
- Worked with teams of 8+ staff to manage lane bookings, food service & events using time management skills

Sports Secretary – Filipino Society of University of Sheffield

Sep 2022 – Jun 2023

- Organised & led 5 sport events including a sports day attended by 100+ members, boosting member engagement by 10%
- Managed a £500 budget covering equipment, venue sourcing and event coordination across all society sport activities.
- Improved team coordination through meetings, shared workspaces & task lists to manage tasks & planning across events.

Skills, Achievements & Interests

Skills: Python, JavaScript, HTML, CSS, SQL, PHP, NumPy, Matplotlib, Pandas, Scikit-learn, Tkinter, Seaborn, Tableau

Tools: Git, Docker, MySQL, SQLite, LAMP stack

Achievements: Varsity Basketball Champions 2024, 1st Dan Black Belt.

Interests: Basketball (Competitive), Boxing, Volleyball, Karate, Gym, Cooking.

References available upon request